Exact Finite Costs and the Non-Gaussian Contraction Limit of Randomized Quicksort

Route-A source-local certificate HCS-C302

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Abstract

We determine the comparison-cost distribution of classical single-pivot Quicksort by an exact probability-generating-polynomial recurrence and derive closed formulas for its mean and variance at every input size. After exact centering, the normalized costs converge in quadratic Wasserstein distance to the unique centered finite-variance fixed point of the Quicksort distributional transform. Its variance is $7 - 2\pi^2/3$ and its positive third moment is $16\zeta(3) - 19$, proving a genuinely non-Gaussian limit; endpoint pivots, cost conventions and Route-A boundaries are closed explicitly.

1 The finite recursive law

Let X_n count key comparisons made by classical single-pivot Quicksort on a uniform random permutation of n distinct keys. The pivot is compared once with every other key, and $X_0 = X_1 = 0$. If I_n is uniform on $\{0, \dots, n-1\}$, the relative orders in the two subarrays are independent uniform permutations; hence

$$X_n \overset{d}{=} X_{I_n} + X'_{n-1-I_n} + n - 1. \tag{1}$$

All variables on the two recursive branches are independent conditional on I_n.

Theorem 1 (Every finite law and its first two moments)**.** *For $G_n(z) = \mathbb{E}z^{X_n}$, $G_0 = G_1 = 1$ and*

$$G_n(z) = \frac{z^{n-1}}{n} \sum_{j=0}^{n-1} G_j(z) G_{n-1-j}(z), \qquad n \geq 2. \tag{2}$$

Writing $H_n = \sum_{k=1}^{n} k^{-1}$ and $H_n^{(2)} = \sum_{k=1}^{n} k^{-2}$, with empty sums zero,

$$\mu_n := \mathbb{E}X_n = 2(n+1)H_n - 4n, \tag{3}$$

$$v_n := \operatorname{Var} X_n = 7n^2 - 4(n+1)^2 H_n^{(2)} - 2(n+1)H_n + 13n. \tag{4}$$

These formulas include $n = 0, 1$.

Proof. Conditioning (1) on the pivot rank and multiplying independent PGFs gives (2). Expectations satisfy

$$\mu_n = n - 1 + \frac{2}{n} \sum_{j=0}^{n-1} \mu_j, \tag{5}$$

whose successive difference is solved by (3). Total variance gives

$$v_n = \frac{2}{n} \sum_{j=0}^{n-1} v_j + \operatorname{Var}(\mu_{I_n} + \mu_{n-1-I_n}). \tag{6}$$

Substitution of (3), followed by the elementary harmonic-sum identities, reduces (6) to (4). Both formulas also verify the two boundary values directly. □

2 The contraction limit

Put $Y_n = (X_n - \mu_n)/(n+1)$ and define, with $0\log 0 = 0$,

$$C(u) = 1 + 2u\log u + 2(1-u)\log(1-u). \tag{7}$$

Theorem 2 (Unique fixed point and L^2 convergence)**.** *The laws of Y_n converge in quadratic Wasserstein distance, and under a recursive coupling in L^2, to the unique centered finite-second-moment law satisfying*

$$Y \overset{d}{=} UY_1 + (1-U)Y_2 + C(U), \tag{8}$$

where U is uniform on $[0,1]$ and Y_1, Y_2 are independent copies of Y, independent of U.

Proof. Center (1) using (3). With $A_n = (I_n+1)/(n+1)$, $B_n = 1 - A_n$, the exact recurrence is

$$Y_n = A_n Y_{I_n} + B_n Y'_{n-1-I_n} + C_n(I_n), \quad C_n(j) = \frac{n-1+\mu_j+\mu_{n-1-j}-\mu_n}{n+1}.$$

Couple $I_n = \lfloor nU \rfloor$. Then $A_n \to U$ uniformly. Standard harmonic bounds, with the two endpoints evaluated directly, give $C_n(\lfloor nU \rfloor) \to C(U)$ in L^2; the exact mean recurrence gives $n^{-1}\sum_j C_n(j) = 0$, hence $\int_0^1 C(u)\,du = 0$.

On centered finite-second-moment laws, couple two inputs optimally and use independent copies on the two branches. Cross terms vanish and the transform T in (8) obeys

$$d_2(T\nu, T\nu')^2 \le \mathbb{E}\{U^2 + (1-U)^2\} d_2(\nu,\nu')^2 = \frac{2}{3} d_2(\nu,\nu')^2. \tag{9}$$

Banach's theorem gives one fixed law. On an i.i.d.-U binary tree, give node v the product weight L_v along its root path and put $\Delta_r = \sum_{|v|=r} L_v C(U_v)$. Distinct levels are orthogonal and $\mathbb{E}\Delta_r^2 = (\mathbb{E}C(U)^2)(2/3)^r$, so $\sum_r \Delta_r$ converges in L^2; splitting at the root realizes the unique fixed law. On the same tree realize $\widehat{Y}_n$ by using $I_m = \lfloor mU_v \rfloor$ whenever a node has size m. Then $\widehat{Y}_n =^d Y_n$. Put $e_n = \|\widehat{Y}_n - Y\|_2$. Centering and independence of the two subtrees, followed by Minkowski's inequality, give

$$e_n \le Q_n^{1/2} + \delta_n, \qquad Q_n = \frac{2}{n}\sum_{j=0}^{n-1}\left(\frac{j+1}{n+1}\right)^2 e_j^2, \qquad \delta_n \to 0. \tag{10}$$

Here δ_n is the L^2 norm of the coefficient and grid-toll error; the uniform coefficient error is $O(n^{-1})$, and the harmonic estimate above makes the toll part $o(1)$. Formula (4) makes (e_n) bounded. Put $M_N = \sup_{j\ge N} e_j$, so $M_N \downarrow D = \limsup e_n$, and split Q_n at this cutoff. The finitely many small-j terms have total weight $O(n^{-3})$; the rest is bounded by M_N. Since

$$\frac{2}{n}\sum_{j=0}^{n-1}\left(\frac{j+1}{n+1}\right)^2 = \frac{2n+1}{3(n+1)} \longrightarrow \frac{2}{3},$$

the limsup of (10) gives $D \le \sqrt{2/3}\, M_N$. Letting $N \to \infty$ yields $D \le \sqrt{2/3}\, D$, which forces $D = 0$. Thus this explicit recursive coupling realizes L^2 convergence and, in particular, the laws converge in d_2. □

3 Exact moments and non-Gaussianity

Lemma 1 (Third-moment license)**.** *The fixed law in Theorem 2 has a finite absolute third moment.*

Proof. On an independent binary tree set $L_\varnothing = 1$, $L_{v0} = L_v U_v$, $L_{v1} = L_v(1-U_v)$, and $\Delta_r = \sum_{|v|=r} L_v C(U_v)$. The toll is bounded and centered. A conditional Rosenthal inequality at level r, together with

$$\mathbb{E}\sum_{|v|=r} L_v^2 = (2/3)^r, \qquad \mathbb{E}\sum_{|v|=r} L_v^3 = (1/2)^r, \qquad \sum_{|v|=r} L_v^2 \le 1,$$

gives $\|\Delta_r\|_3 \le K\{(2/3)^{r/3} + (1/2)^{r/3}\}$. Thus $\sum_r \Delta_r$ converges in L^3. Splitting it at the root gives (8); its centered finite-variance law is the unique L^2 fixed point, proving the claim. □

Taking second moments in (8), or taking the leading coefficient in (4), gives

$$m_2 = \mathbb{E}Y^2 = 7 - \frac{2\pi^2}{3} > 0. \tag{11}$$

Lemma 1 licenses cubing the fixed-point identity. Independence and centering give

$$m_3 = \frac{1}{2}m_3 + 3m_2\int_0^1 C(u)(u^2+(1-u)^2)\,\mathrm{d}u + \int_0^1 C(u)^3\,\mathrm{d}u. \tag{12}$$

Here the two integrals are respectively $1/18$ and $-32/3+\pi^2/9+8\zeta(3)$. Beta-function differentiation then evaluates (12) as

$$\mathbb{E}Y^3 = 16\zeta(3) - 19 > 0. \tag{13}$$

Moreover $\zeta(3) > \sum_{k=1}^{6} k^{-3} = 28567/24000$, so the displayed moment exceeds $67/1500$. A centered Gaussian has zero third moment, so the limit is not Gaussian.

4 Boundary, evidence, and Route A

Empty and singleton inputs cost zero; two keys cost one. Extreme pivots, including an empty branch, are already present in (2). The frozen cost counts comparisons only: swaps, recursion depth, repeated keys, three-way partitioning and sampled pivots are separate models. Normalization by n has the same asymptotic law for $n \geq 1$ but is not the exact finite recurrence used here.

The evidence artifact expands every PGF through a declared cutoff, recomputes moments from coefficients, verifies the harmonic formulas and beta derivatives independently, and attacks centering, endpoints and a false Gaussian claim. Finite rows are regression evidence; the all-n theorem is the analytic recurrence and contraction argument.

C291's first-event convolution belongs to dimer adsorption, not recursive permutation splitting. The strict Route-A tuple is (A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL) and Route B is locked. Input size is not an arithmetic clock, finite PGFs are not target determinants, and the distributional fixed point is not a Hilbert–Pólya operator. Scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

Literature ownership and AI-use statement

Hoare introduced Quicksort [1]; Régnier [2] and Rösler [3] established its limiting distribution and contraction approach. We make no literary priority claim. AI tools assisted exact-algebra checks, hostile review and manuscript preparation; all formulas and proof obligations are exposed here and in the independent certificate.

References

[1] C. A. R. Hoare, "Quicksort," *The Computer Journal* 5 (1962), 10–16, doi:10.1093/comjnl/5.1.10.

[2] M. Régnier, "A limiting distribution for quicksort," *RAIRO Theor. Inform. Appl.* 23 (1989), 335–343.

[3] U. Rösler, "A limit theorem for Quicksort," *RAIRO Theor. Inform. Appl.* 25 (1991), 85–100.